

Presentation Information

Language

The working language of the workshop is English, which includes all presentations, printed materials, and discussions.

Presentation Length and Format

All presentations are oral and are allotted 20-minute time slots, which includes 15 minutes for the presentation and 5 minutes for questions and discussion.

All presenters should contact the session chairs no later than 15 minutes before the start of the session to confirm their presence and upload their presentation files to the session computer. If the session starts immediately after the end of the previous session, presenters should arrive to the session room and contact the session chairs during the coffee break or lunch break preceding the start of the session.

Technical Equipment

Each session room is equipped with a laptop computer connected to a video projector. Video resolution is 1280×800 pixels. Session room computers run under Windows 7 Pro and have Microsoft Office 2010 installed.

File Formats

The only file formats supported by the workshop computers are Microsoft PowerPoint (*.ppt, *.pptx) and Adobe PDF (*.pdf).

File Upload

Speakers are asked to upload their presentation files to the session room computer no later than 15 minutes before the session start. If the session starts immediately after the end of the previous session, speakers should upload their files during any breakfast, coffee, or lunch break preceding the session (or even the day/evening before).

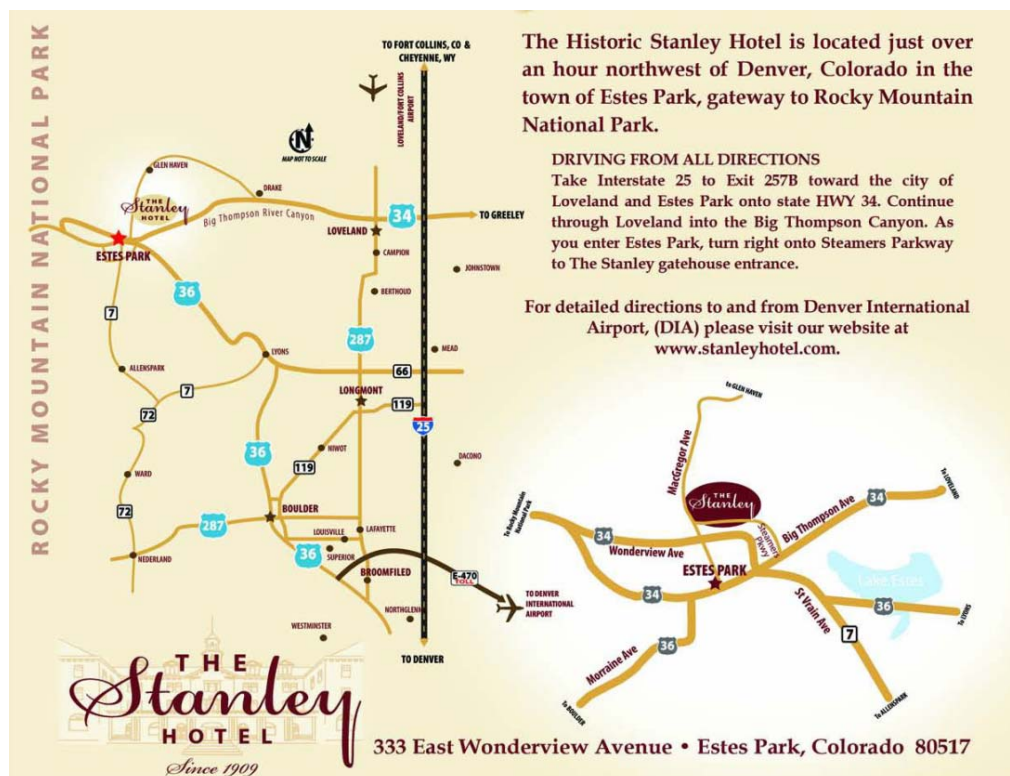
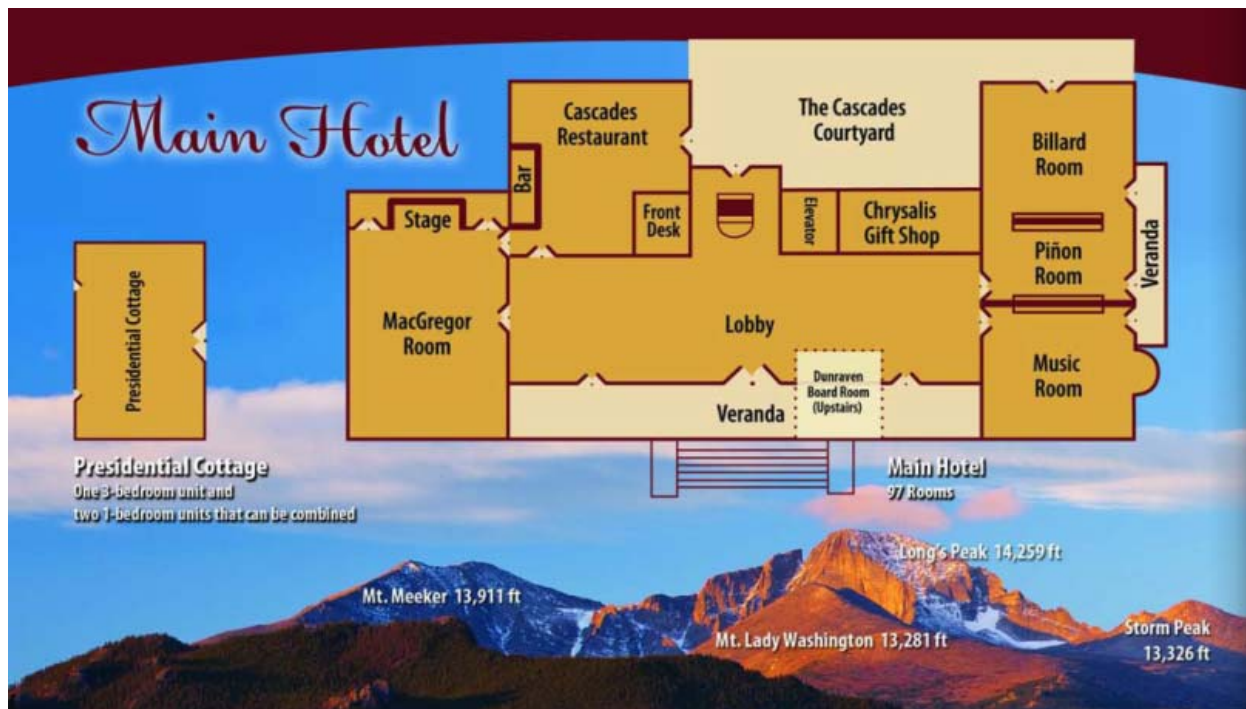
Session room computers are equipped with USB ports and DVD drives. Staff personnel will be available to assist with the upload procedure.

Session Chairs

Special session organizers will serve as session chairs. Session chairs must arrive to the session room no later than 15 minutes before the session start. If the session starts immediately after the end of the previous session, session chairs should come to the room and check if everything is ready during the coffee break or lunch break preceding the start of the session.

Conference Floor Plan and Hotel Directions

Sessions are held in Billiard Room and Music Room, registration and coffee breaks are in Pinion Room, breakfasts, lunches, and conference banquet are in MacGregor Room.





Detailed Technical and Social Program

Sunday, June 3, 2012		
19:00- 21:00	Welcome reception (Music Room)	
Monday, June 4, 2012		
7:00	Breakfast (MacGregor Room)	
	Billiard Room	Music Room
8:00	Session 1: Modeling and Design of Antennas and Arrays Using FEM (R. Kindt, D. Filipovic, E. Topsakal) 1.1. Highly Coupled Bow-Tie Array Design with Passive Elements <i>Safa Salman*, Dimitris Psychoudakis, and John L. Volakis</i>	Session 2: Discontinuous Galerkin Methods (Q. H. Liu, W. Cai) 2.1. On the Physics, Accuracy, and Efficiency of the Dual-Field Domain-Decomposition Method for Time-Domain Electromagnetic Simulation <i>Xiaolei Li and Jian-Ming Jin*</i>
8:20	1.2. Fully Overlapping Domain Decomposition in Simulating Irregularly Arranged Arrays in Large Media <i>Tao Peng*, Kubilay Sertel, and John L. Volakis</i>	2.2. A Full Vectorial Generalized Discontinuous Galerkin Beam Propagation Method for Optical Waveguides <i>Wei Cai*</i>
8:40	1.3. Modeling of Antennas and Arrays Using Domain Decomposition Method <i>Kezhong Zhao* and Nancy Lambert</i>	2.3. Non-Conformal and Parallel and Interior Penalty Discontinuous Galerkin Method for the Time-Domain Maxwell's Equations <i>Stylianios Dosopoulos and Jin-Fa Lee</i>
9:00	1.4. Theoretical Analysis of Varactor-loaded Half-width Leaky-wave Antenna Arrays <i>Leo C. Kempel*, B. Shanker, Edward J. Rothwell, and Prem Chahal</i>	2.4. Discontinuous Galerkin Methods for Ideal MHD Equations <i>Fengyan Li*</i>
9:20	1.5. Non-conformal Domain Decomposition Methods for Modeling Large Finite Antenna Arrays <i>Zhen Peng, Kheng-Hwee Lim, and Jin-Fa Lee</i>	2.5. New Spectral-Prism-Element for Layered Structures implemented in an efficient implicit DG-FETD method <i>Luis E. Tobon and Qing H. Liu*</i>
9:40	Coffee break (Pinion room)	
10:10	1.6. Investigation of the Scattering Behavior of Transponder Antennas in Case of Nonlinear Termination <i>Thomas Bauernfeind*, Kurt Preis, Gergely Koczka, and Oszkar Biro</i>	2.6. An Efficient Discontinuous Galerkin Finite-Element Time-Domain Method and its Application to Micro-Resistivity Imager <i>Jiefu Chen, Carlos Haramboure, Lance Pate, Shawn Wallace, Jay Martin, Medhat Mickael, and Mac Wisler</i>
10:30	1.7. On the Design of Finite Planar Ultrawideband Modular Antenna (PUMA) Arrays using Domain Decomposition Methods <i>Steve S. Holland, Jack T. Logan, Rick W. Kindt, and Marinos N. Vouvakis*</i>	Session 4: FEM and CEM Applications in Optics and Nanophotonics (E. Simsek) 4.1. Electromagnetic Scattering of Finite Size Periodic Large Scale System: Simulation by Domain Decomposition Spectral Element Method <i>Ma Luo and Qing Huo Liu*</i>
10:50	Session 3: Time-Domain FEM and Applications (D. White, O. Biro, T. Rylander) 3.1. Numerical Simulation of HPM Devices with ICEPIC <i>Andrew D. Greenwood*</i>	4.2. Efficient alternatives to edge vector finite elements for EM field calculations using Hermite interpolation polynomial basis functions on a nodal grid in 2D <i>J. D. Albrecht, C. R. Boucher, C. I. Ahheng, and L. R. Ram-Mohan</i>

Monday, June 4, 2012		
	Billiard Room	Music Room
11:10	3.2. Time Domain Method for Periodic Nonlinear Eddy Current Problems <i>Oszkár Bíró, Gergely Koczka, and Kurt Preis</i>	4.3. Parallel Finite Element Electromagnetic Code Suite for Accelerator Modeling and Simulation <i>Cho Ng, Lixin Ge, Kwok Ko, Kihwan Lee, Zenghai Li, Greg Schussman, and Liling Xiao</i>
11:30	3.3. Hybrid FETD/FDTD Techniques Based on the Discontinuous Galerkin Method <i>Bao Zhu, Jiefu Chen, Wanxie Zhong, and Qing Huo Liu</i>	4.4. Optical, Electromagnetics, and Thermal Modeling of Interaction of a Focused Beam of Light with Plasmonic Nanoparticles <i>Eren S. Unlu and Kursat Sendur*</i>
11:50	Lunch break (MacGregor Room)	
13:10	3.4. Transient Thermal Analysis using a Non-conformal Domain Decomposition Approach <i>Yang Shao*, Zhen Peng, and Jin-Fa Lee</i>	4.5. Finite Element Analysis of Wideband Nanostructures for Photovoltaic Applications <i>Niccolo Breda, Giuseppe Pelosi, Stefano Selleri, and Ruggero Taddei</i>
13:30	3.5. An Explicit and Unconditionally Stable Time-Domain Finite-Element Method of Linear Complexity <i>Qing He, Duo Chen, and Dan Jiao</i>	4.6. Hybrid Optical Waveguides <i>Ergun Simsek* and Qing H. Liu</i>
13:50	3.6. Conformal PML Modeling in DGTD using Continuous Material Properties <i>Jue Wang*, Zhen Peng, and Jin-Fa Lee</i>	Session 6: Advances in Vector Bases for CEM (R. Graglia, A. Peterson, D. Wilton)
14:10	Session 5: Model Order Reduction Techniques (R. Dyczij-Edlinger, V. de la Rubia) 5.1. Reduced-order Electromagnetic Modeling in the Presence of Uncertainty <i>Juan S. Ochoa and Andreas C. Cangellaris*</i>	6.1. On the Construction of Well-conditioned Hierarchical Bases for Tetrahedral H(curl)-Conforming Nedelec Element <i>Wei Cai* and Jianguo Xin</i>
14:30	5.2. Efficient Model Order Reduction for FEM Analysis of Waveguide Structures and Resonators <i>Fotyga G.*, Kulas L., Nyka K., and Mrozowski M.</i>	6.2. High Order Finite Elements for Computational Physics: An LLNL Perspective <i>Robert N. Rieben*</i>
14:50	5.3. Model Order Reduction for Systems with Geometrical Parameters <i>Kynthia K. Stavarakakis*, Tilmann Wittig, Wolfgang Ackermann, and Thomas Weiland</i>	6.3. FEM-MoM-Diakoptic Analysis of Scatterers with Anisotropic Inhomogeneities Using Hierarchical Vector Bases on Large Curved Elements <i>Ana B. Manić*, Dragan I. Olčan, Milan M. Ilić, and Branislav M. Notaroš</i>
15:10	Coffee break (Pinion room)	
15:40	5.4. Reduced-Order Models for Geometric Parameters <i>Stefan Burgard*, Ortwin Farle, and Romanus Dyczij-Edlinger</i>	6.5. Issues Arising in Connection with Singular Vector Bases <i>Roberto D. Graglia and Andrew F. Peterson*</i>
16:00	5.5. Seamless Reduced Basis Element Method for Waveguide Simulation <i>Yanlai Chen*, Jan S. Hesthaven, and Yvon Maday</i>	6.6. An Orthogonal Prism Vector Bases Based Quadratic Eigenvalue Solver of Linear Complexity for 3-D Electromagnetic Analysis <i>Jongwon Lee, Duo Chen, and Dan Jiao</i>
16:20	5.6. The Reduced Basis Method for Saddle Point Problems <i>Karen Veroy* and Anna-Lena Gerner</i>	6.7. Preliminary Research on the Discontinuous Enrichment Method Based Domain Decomposition Scheme for Solving the Three-Dimensional Vector Curl-Curl Equation <i>Ming-Feng Xue* and Jian-Ming Jin</i>
16:40	5.7. Reliable Fast Frequency Sweep of Microwave Circuits via the Reduced-Basis Method <i>Valentin de la Rubia</i>	6.8. An Iterative Domain Decomposition Method for Solving Wave Propagation Problems <i>Gergely Koczka, Kurt Preis*, Thomas Bauernfeind, and Oszkár Bíró</i>
17:00	5.8. Reduced Order Modeling of High Magnetic Field Magnets <i>C. Daversin, C. Prudhomme, C. Trophime*, and S. Veys</i>	6.9. Evaluating Singular and Near-Singular Integrals on Curvilinear Elements <i>Donald R. Wilton*</i>
17:30- 19:00	Stanley Hotel Ghost & History Tour	
19:30- 22:30	Conference Banquet (MacGregor Room) (the room opens at 19:00)	

Tuesday, June 5, 2012		
7:00	Breakfast (MacGregor Room)	
	Billiard Room	Music Room
8:00	Session 7: FETD Modeling of Complex Media and Structures (R. Lee, F. Teixeira) 7.1. FETD/FDTD with Automatic Hybrid Mesh Generation <i>Shumin Wang*</i>	Session 8 Adaptive FEM, Higher Order Bases, and Advanced FEM Formulations (J. Webb, M. Gavrilovic, M. Salazar Palma, L. G. Castillo) 8.1. Hp-adaption for Computing Scattering Parameters on Tetrahedral Meshes <i>J. P. Webb</i>
8:20	7.2. Study of Numerical Dispersion for Quadrilateral and Triangular Elements Used in the Mixed Finite Element Method <i>Luis E. Tobon and Qing H. Liu*</i>	8.2. Efficient RCS Prediction of Jet Engine Air Intakes Using hp-FEM and Reduced Order Modeling <i>Adam Zdunek* and Waldemar Rachowicz</i>
8:40	7.3. Probabilistic Finite-Difference Time-Domain Method Through Stochastic Electromagnetic Macro-Models <i>Ata Zadehgoi*, Andreas C. Cangellaris</i>	8.3. Rules for Adoption of Expansion and Integration Orders in FEM Analysis Using Higher Order Hierarchical Bases on Generalized Hexahedral Elements <i>Nada J. Šekeljić*, Slobodan V. Savić, Milan M. Ilić, and Branislav M. Notaroš</i>
9:00	7.4. Advancing Ultrahigh Field Human MRI using FDTD Method <i>Tamer S. Ibrahim*</i>	8.4. Auxiliary Space Preconditioning for Hierarchical Elements <i>A. Aghabarati, J. P. Webb</i>
9:20	7.5. The Use of Multiple Time Steps in the FDTD Method <i>Ruben Ortega and Robert Lee*</i>	8.5. Chain Homotopy Operators and Whitney form-based Finite Element Analysis <i>P. Robert Kotiuga*</i>
9:40	Coffee break (Pinion room)	
10:10	Session 9: Advances in Hybrid Methods and Multiphysics Problems (B. Shanker, L. Kempel) 9.2. Thermal-Aware DC IR-Drop Co-Analysis using a Non-conformal Domain Decomposition Approach <i>Yang Shao*, Zhen Peng, Jin-Fa Lee</i>	8.6. Comparative Performance of Nodal-Based Versus Edge-Based Finite Element Formulations <i>Ruben Otin*, Santiago Badia, and Luis E. Garcia-Castillo</i>
10:30	9.3. Development of Time Domain Discontinuous Galerkin-Vector Generalized Finite Element Method <i>O. Tuncer, B. Shanker*, L. C. Kempel</i>	8.8. A Three-Dimensional Self-Adaptive hp Finite Element Method for the Characterization of Waveguide Discontinuities <i>Ignacio Gomez-Revuelto, Luis E. Garcia-Castillo, Sergio Llorente-Romano, and David Pardo</i>
10:50	9.4. Multi-solver-based Generalized Impedance Boundary Condition for Complicated EM Simulation <i>Shiquan He, Jun Hu*, Zaiping Nie, Lijun Jiang, W. C. Chew</i>	Session 10: FEM Modeling and Applications of Metamaterials and Periodic Media (K. Sertel, J. Volakis) 10.1. Finite Element Modeling of Magnetic Metamaterials and Associated Challenges for Their Experimental Characterization <i>Nil Apaydin*, Kubilay Sertel, and John L. Volakis</i>
11:10	9.5. Analysis of Electromechanical Devices Using the Dual-Primal Finite Element Tearing and Interconnecting Method Incorporated with the LU Recombination Method <i>Wang Yao*, Jian-Ming Jin</i>	10.2. A Method for Rapid Evaluation of Space-time Convolutions with the Time Domain Periodic Green's Function <i>D. Dault and B. Shanker*</i>
11:30	9.6. Alternative Formulations for Hybrid Electromagnetic-Circuit Simulators <i>Vivek Subramanian, Ali E. Yilmaz*</i>	10.3. Finite Element Analysis of Photon Density of States for Two-Dimensional Photonic Crystals with Omnidirectional Light Propagation <i>R. F. Jao and M. C. Lin</i>
11:50	Lunch break (MacGregor Room)	
13:10	9.7. Symmetric and Nonsymmetric FEM-MoM Techniques Using Higher Order Hierarchical Vector Basis Functions and Curved Parametric Elements <i>Ana Manić*, Milan Ilić, Branislav Notaroš</i>	10.4. A Hybrid Approach for Characterizing Large Anisotropic and Inhomogeneous Metamaterial Structures <i>Justin G. Pollock and Ashwin K. Iyer*</i>

Tuesday, June 5, 2012		
	Billiard Room	Music Room
13:30	Session 11: Domain-Decomposition Methods (J.-F. Lee, Z. Peng) 11.1. Analysis of Three-Dimensional Array Structures Using Nonconformal and Cement FETI-DP Methods <i>Ming-Feng Xue* and Jian-Ming Jin</i>	Session 12: Advanced FEM/MoM Modeling, Design, and Optimization (C.J. Reddy, U. Jakobus, J. Zapata, J. Gil, D. Jiao, A. Cangellaris) 12.1. Overview of the Hybrid Finite Element Method/Method of Moments Capability in FEKO <i>Marianne Bingle*, Ulrich Jakobus, and Johann J. van Tonder</i>
13:50	11.2. A Domain-Decomposition-Based Preconditioner of FE-BI-MLFMA for Computing EM Scattering by Large Inhomogeneous 3D Targets <i>Ming-Lin Yang, Xin-Qing Sheng, and Cheng-Dan Huang</i>	12.2. Application of Hybrid FEM/MoM Technique for Handheld Mobile Devices <i>Rensheng Sun and C. J. Reddy</i>
14:10	11.3. On the Computational Complexity of Domain Wise Multilevel Multifrontal Method <i>Peng Liu* and Chao-Fu Wang</i>	12.3. Application of Hybrid FEM/MoM Technique for Cavity Backed Apertures <i>Gopinath Gampala, Rensheng Sun and C. J. Reddy</i>
14:30	11.4. Harmonic Balance Domain Decomposition Finite Element for Nonlinear Passive Microwave Devices Analysis <i>Laurent Ntibarikure, Giuseppe Pelosi, and Stefano Selleri</i>	12.4. Eigenproblem Approach for Analysis and Optimization of Microwave Filters Using Finite Element Method <i>Adam Lamecki, Michal Mrozowski</i>
14:50	11.5. Application of Tree-Cotree Splitting to the Dual-Primal Finite Element Tearing and Interconnecting Method for Solving Low-Frequency Breakdown Problems <i>Wang Yao* and Jian-Ming Jin</i>	12.5. Does Sensitivity Analysis for Fast Frequency Sweeps Converge? <i>Andreas Köhler, Ortwin Farle, Romanus Dyczij-Edlinger*</i>
15:10	Coffee break (Pinion room)	
15:40	11.6. Edge Elements and DPM with PBSV to Accelerate Solving Scattering Problems of Electrically Large Objects <i>Lianyou Sun*, Lishan Xue, and Wei Hong</i>	12.6. Benchmark of Conformal Finite-Difference Method against Finite Element Method <i>M. C. Lin</i>
16:00	11.7. Enhancement of Second-harmonic Generation in an Air-bridge Photonic Crystal Slab: Simulation by Spectral Element Method <i>Ma Luo and Qing Huo Liu*</i>	12.7. Layout-Integrated Electromagnetic Interconnect Characterization and Simulation <i>Dennis Nagle, Jilin Tan, Jian-Ming Jin</i>
16:20	11.8. A Non-overlapping and Non-conformal Domain Decomposition Method with Optimized Second Order Transmission Conditions for Time-Harmonic Maxwell Equations in R^3 <i>Zhen Peng and Jin-Fa Lee</i>	12.8. An Extraction-Free Circuit Simulator of Linear Complexity Guided by Electromagnetics-Based First Principles <i>Qing He, Duo Chen, and Dan Jiao</i>
16:40	11.9. The Quest for Robust Domain Decomposition Methods in CEM: Are We There Yet? <i>Marinos N. Vouvakis* and Georgios N. Paraschos</i>	12.9. A model reduction method for multiple particle electromagnetic scattering in three dimensions <i>M. Ganesh*, J. Hesthaven, and B. Stamm</i>
17:00	Technical program ends for the day	
17:00- 18:30	FEM2012 Scientific Committee and Organizing Committee Meeting (Music Room)	
17:30- 19:00	Stanley Hotel Ghost & History Tour	

Wednesday, June 6, 2012		
7:00- 8:00	Breakfast (MacGregor Room)	
8:00- 14:00	Trip to Rocky Mountain National Park (picnic lunch)	
	Open Forum Discussions (Music Room)	
15:00	Open forum discussion 1: Open Forum Discussion on FEM and CEM Bases, Elements, and Formulations (A. Peterson, D. Wilton, J. Webb, K. Sertel)	
15:45	Open forum discussion 2: Open Forum Discussion on DDM, MOR, and Efficient FEM and Hybrid Solutions (J.-F. Lee, Zhen Peng, R. Dyczij-Edlinger, V. de la Rubia)	
16:30	Coffee break (Pinion Room)	
16:50	Open forum discussion 3: Open Forum Discussion on FEM-Based Modeling, Design, and Applications (L. Kempel, B. Shanker, R. Kindt, C.J. Reddy, K. Zhao, T. Euler)	
18:00- 21:00	Closing Reception (The Elkhorn Lodge & Ranch - Chuckwagon Dinner and Cowboy Show)	